

23)2D array coloumn sum:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
void ans(int **arr,int n){
```

```
    for(int i=0;i<n;i++){
```

```
        int sum=0;
```

```
        for(int j=0;j<n;j++){
```

```
            sum+=arr[j][i];
```

```
        }
```

```
        printf("%d \n",sum);
```

```
    }
```

```
}
```

```
int main()
```

```
{
```

```
    int n;
```

```
    scanf("%d",&n);
```

```
    int **arr=(int**)malloc(n*sizeof(int*));
```

```
    for(int i=0;i<n;i++){
```

```
        *(arr+i)=(int*)malloc(n*sizeof(int));
```

```
    }
```

```
    for(int i=0;i<n;i++){
```

```
        for(int j=0;j<n;j++){
```

```
            scanf("%d",&arr[i][j]);
```

```
        }
```

```
    }
```

```
    ans(arr,n);
```

```
    return 0;
```

```
}
```

24)2D array max element in each row:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
void ans(int **arr,int n){
```

```
    for(int i=0;i<n;i++){
```

```
        int max=arr[i][0];
```

```
        for(int j=0;j<n;j++){
```

```
            if(max<arr[i][j]){
```

```
                max=arr[i][j];
```

```
            }
```

```
        }
```

```
        printf("%d \n",max);
```

```
    }
```

```
}
```

```
int main()
```

```
{
```

```
    int n;
```

```
    scanf("%d",&n);
```

```
    int **arr=(int**)malloc(n*sizeof(int*));
```

```
    for(int i=0;i<n;i++){
```

```
        *(arr+i)=(int*)malloc(n*sizeof(int));
```

```
    }
```

```
    for(int i=0;i<n;i++){
```

```
        for(int j=0;j<n;j++){
```

```
            scanf("%d",&arr[i][j]);
```

```
        }
```

```
    }
```

```
    ans(arr,n);
```

```
    return 0;
```

```
}
```

25)2D array max element in each coloumn:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
void ans(int **arr,int n){
```

```
    for(int i=0;i<n;i++){
```

```
        int max=arr[0][i];
```

```
        for(int j=0;j<n;j++){
```

```
            if(max<arr[j][i]){
```

```
                max=arr[j][i];
```

```
            }
```

```
        }
```

```
        printf("%d \n",max);
```

```
    }
```

```
}
```

```
int main()
```

```
{
```

```
    int n;
```

```
    scanf("%d",&n);
```

```
    int **arr=(int**)malloc(n*sizeof(int*));
```

```
    for(int i=0;i<n;i++){
```

```
        *(arr+i)=(int*)malloc(n*sizeof(int));
```

```
    }
```

```
    for(int i=0;i<n;i++){
```

```
        for(int j=0;j<n;j++){
```

```
            scanf("%d",&arr[i][j]);
```

```
        }
```

```
    }
```

```
    ans(arr,n);
```

```
    return 0;
}
```

26)2D array overall max element:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
void ans(int **arr,int n){
    int max=arr[0][0];
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            if(max<arr[i][j]){
                max=arr[i][j];
            }
        }
    }
    printf("%d\n",max);
}

int main()
{
    int n;
    scanf("%d",&n);
    int **arr=(int**)malloc(n*sizeof(int*));
    for(int i=0;i<n;i++){
        *(arr+i)=(int*)malloc(n*sizeof(int));
    }
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            scanf("%d",&arr[i][j]);
        }
    }
}
```

```

    }
    ans(arr,n);
    return 0;
}

```

27) Transpose of a Matrix:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```

void ans(int **arr,int n){
    int copy[n][n];
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            copy[j][i]=arr[i][j];
        }
    }
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            printf("%d ",copy[i][j]);
        }
        printf("\n");
    }
}

```

```

}

int main()
{
    int n;
    scanf("%d",&n);
    int **arr=(int**)malloc(n*sizeof(int*));
    for(int i=0;i<n;i++){
        *(arr+i)=(int*)malloc(n*sizeof(int));
    }
}

```

```

for(int i=0;i<n;i++){
    for(int j=0;j<n;j++){
        scanf("%d",&arr[i][j]);
    }
}
ans(arr,n);
return 0;
}

```

28) Primary Diagonal Sum:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```

void ans(int **arr,int n){
    int sum=0;
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            if(i==j){
                sum+=arr[i][j];
            }
        }
    }
    printf("%d",sum);
}

```

```
}
```

```
int main()
```

```
{
```

```
    int n;
```

```
    scanf("%d",&n);
```

```
    int **arr=(int**)malloc(n*sizeof(int*));
```

```
    for(int i=0;i<n;i++){
```

```

        *(arr+i)=(int*)malloc(n*sizeof(int));
    }
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            scanf("%d",&arr[i][j]);
        }
    }
    ans(arr,n);
    return 0;
}

```

29)Secondary diagonal sum:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
void ans(int **arr,int n){
```

```
    int sum=0,k=n-1;
```

```
    for(int j=0;j<n;j++){
```

```
        sum+=arr[k][j];
```

```
        k--;
```

```
    }
```

```
    printf("%d",sum);
```

```
}
```

```
int main()
```

```
{
```

```
    int n;
```

```
    scanf("%d",&n);
```

```
    int **arr=(int**)malloc(n*sizeof(int*));
```

```
    for(int i=0;i<n;i++){
```

```
        *(arr+i)=(int*)malloc(n*sizeof(int));
```

```

    }
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            scanf("%d",&arr[i][j]);
        }
    }
    ans(arr,n);
    return 0;
}

```

30)row col zero transformation:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
void ans(int **arr,int n ,int m){
```

```
    int row[n],col[m],x=0,y=0;
```

```
    for(int i=0;i<n;i++){
```

```
        for(int j=0;j<m;j++){
```

```
            if(arr[i][j]==0){
```

```
                row[x]=i;
```

```
                col[y]=j;
```

```
                x++;
```

```
                y++;
```

```
            }
```

```
        }
```

```
    }x--;
```

```
    y--;
```

```
    while(x>=0){
```

```
        for(int j=0;j<m;j++){
```

```
            arr[row[x]][j]=0;
```

```
        }
```



```

x--;

for(int i=0;i<n;i++){
    arr[i][col[y]]=0;
}
y--;
}
}

int main()
{
    int n,m;
    scanf("%d %d",&n,&m);
    int **arr=(int**)malloc(n*sizeof(int*));
    for(int i=0;i<m;i++){
        *(arr+i)=(int*)malloc(m*sizeof(int));
    }
    for(int i=0;i<n;i++){
        for(int j=0;j<m;j++){
            scanf("%d",&arr[i][j]);
        }
    }
    ans(arr,n,m);
    for(int i=0;i<n;i++){
        for(int j=0;j<m;j++){
            printf("%d ",arr[i][j]);
        }
        printf("\n");
    }
    return 0;
}

```